Gustavo Landfried
Chemin des Bégonias 1
1018 Lausanne

Abril de 2026

Madame, Monsieur,

Soy ciudadano suizo radicado en Lausanne, doctor en Ciencias de la Computación con especialización en inferencia bayesiana causal y licenciado en Ciencias Antropológicas. Me instalé en la Suiza romanda en abril de 2026 tras varios años en Argentina, donde desarrollé mi actividad de investigación, docencia y consultoría. Trabajo actualmente como Senior Data Scientist en modalidad remota para multinacionales latinoamericanas. Esta combinación de formaciones no es accidental: es una trayectoria construida deliberadamente para operar en la intersección entre el rigor estadístico y la comprensión de fenómenos sociales, económicos y tecnológicos.

En los últimos años he liderado proyectos estadísticos con múltiples interlocutores institucionales. Para la *Foundation for Innovative New Diagnostics* (FIND) y Fiocruz, conduje la evaluación estadística de tests diagnósticos de Chagas en cuatro países mediante modelos B-LCA; en el proceso, detecté un error metodológico en el consorcio europeo COST Action CA18208, lo que derivó en una publicación en *PLoS Neglected Tropical Diseases* (2024). He desarrollado modelos bayesianos jerárquicos para decisiones de política comercial en Coca-Cola FEMSA y KAVAK. Mis publicaciones en *Journal of Statistical Software* (2025), *Journal of Computational Social Science* (2025) y *PLoS ONE* (2019) reflejan la capacidad de combinar rigor metodológico con resultados comunicables. Manejo con fluidez R (Stan), Python (PyMC, Pyro, PyTorch), Julia (Turing) y arquitecturas de datos a escala (Delta Lakehouse, Spark).

Como docente universitario de Inferencia Bayesiana Causal (UNSAM/UBA, 2024–actual), he desarrollado un programa propio de ocho unidades que cubre desde la especificación de modelos gráficos causales hasta la toma de decisiones ergódica. Como fundador y coordinador de Bayes Plurinacional, organizo eventos científicos anuales en múltiples países de América Latina (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). He dirigido 7 tesis de licenciatura en Computación y Ciencia de Datos en la Universidad de Buenos Aires.

La inferencia bayesiana causal es el enfoque que se necesita para cuantificar incertidumbre y separar correlaciones de efectos causales genuinos. Tiene aplicación directa en múltiples dominios: salud pública, estadística y prospectiva institucional, decisiones de negocio con datos escasos, investigación académica y formación especializada. Busco en la Suiza romanda porque es la región donde me instalo con mi esposa (nos casamos en Lausanne hace 15 años) y donde queremos construir nuestra vida de manera duradera. Estoy disponible de manera inmediata, sin necesidad de reubicación.

Atentamente,

LANDFRIED

Gustavo Landfried

Ciencia de datos causal

Landfried, Gustavo

Nacionalidad	Argentino - Suizo
Idiomas	Español nativo | Inglés C1 | Francés C1 | Alemán A1
Residencia	Lausanne (Suiza) | Buenos Aires (Argentina)
Nacimiento	Octubre 1985
Teléfono	+41 77 407 1874 | +54 911 5103 7334
Email	gustavolandfried@gmail.com
Web	glandfried.github.io/
Linkedin	linkedin.com/in/gustavo-landfried
Github	github.com/glandfried

Doctor en Computación. Antropólogo social. **Inferencia Bayesiana Causal**.

- Responsable de proyectos estadísticos en organismos y corporaciones internacionales
- Investigador con publicaciones en inferencia bayesiana causal (JSS, JCSS, PLoS ONE)
- Fundador y coordinador de la comunidad bayesiana de América Latina, Bayes Plurinacional
- Desarrollador de software estadístico a medida de cada problema (Pyro, PyMC)
- Especialista en métodos estadísticos eficientes para la detección de variables ocultas
- Flexibilidad interdisciplinaria y comunicación accesible para audiencias diversas

Profesional estadístico en organismos y corporaciones internacionales

2025 – actual

Causal Data Scientist @ *América Latina*

○ **Coca-Cola (FEMSA)**. Análisis de efectos causales heterogéneos en el sistema de promociones de fidelización Juntos+. Problema: las promociones se desarrollaban por reglas fijas. Resultado: desarrollo de la estrategia adoptada por el equipo, basada en una red bayesiana causal.
○ **Marketplace automotriz (KAVAK)**. Evaluación de impacto de los esfuerzos de marketing. Problema: AB Testing con baja significancia estadística y atribución por regla fija. Resultado: desarrollo una metodología eficiente en datos para el análisis de experimentos causales y atribución Data-Driven.
○ **Optimizador de paid media (Muttdata)**. Distribución de presupuesto entre canales publictarios. Problema: mal desempeño predictivo para conversiones de calidad (baja frecuencia). Resultado: desarrollo de un modelo bayesiano jerárquico que mejoró significativamente la predicciones de impacto.

2023 – 2024

Responsable de la evaluación de test diagnósticos @ *América Latina y Europa*

○ **Foundation for Innovative New Diagnostics (FIND)**. Evaluación de test diagnóstico rápido de Chagas en Sudamérica. Problema: la metodología actual, costosa y lenta, es un cuello de botella. Resultado: mostramos que los test rápidos permiten desarrollar una metodología más eficiente y segura.
○ **European Cooperation in Science and Technology (Action CA18208)**. "Novel tools for test evaluation and disease prevalence estimation". Problema: reporté al consorcio Europeo un error que estaban cometiendo en el uso del modelo Dendukuri. Resultado: publicación en *Plos Neglected Tropical Disease*.
○ **Administración Nacional de Salud (Malbrán).** Experimento de impacto de la cepas de Chagas en los diagnósticos. Problema: baja significancia estadística debido a pocos datos. Resultado: alta significancia mediante modelo bayesiano causal. Publicación en *Journal of Molecular Diagnostics*.

2022 – actual

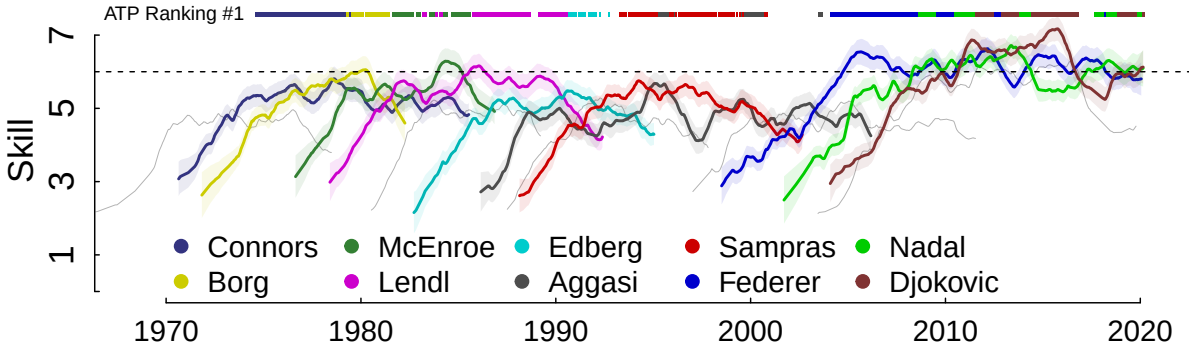
Promotor de la comunidad continental de estadística bayesiana @ *América Latina*

○ **Bayes Plurinacional**. Problema: la adopción de la inferencia Bayesiana está retrasada a nivel continental. Resultado: organización ininterrumpida de eventos de Inteligencia Artificial y formación de recursos humanos en estadística bayesiana, Santiago 2023, Salta 2024, Bogotá 2025, Ciudad Del Este 2026.
○ **Formación profesional en Inferencia Bayesiana Causal**. Separar la correlación de la causalidad y encontrar las decisones óptima a través del tiempo. Contenido completo en Enseñanza.

Sistema de razonamiento óptimo en contextos de incertidumbre

2021 – 2022 Estimador de habilidad estado-del-arte @ *Journal of Statistical Software*

Inferencia eficiente en redes causales con millones de nodos y estructuras irregulares a través de algoritmos de pasaje de mensajes y métodos de aproximación analítica.



2019 – 2020 Impacto de la red social sobre la habilidad @ *Journal of Computational Social Sicence*

Estudio de impacto que la estructura social tienen sobre la habilidad de los individuos. Problema: estudios socio-culturales clásicos no podían evaluar el impacto de la estructura social en el conocimiento humano. Resultado: datos masivos de la comunidad OGS (Online Go Plataform), análisis de grafos (popularity, closeness, intermediacy) y el estimador de habilidad estado-del-arte, permitieron evaluar el impacto.

2017 – 2018 Impacto de la formación de equipos en la habilidad @ *PLoS ONE*

Analizamos una comunidad del juego Risk, entre el año 2006 y el año 2013, en la que se podía jugar en equipo. Mostramos que jugar en equipo conduce a una habilidad más alta a largo plazo, y que jugar con compañero estable acelera la adquisición de habilidades.

Profesional en antropología social con orientación cuantitativa.

2015 – 2016 Programa de Análisis Social de la Ciudadanía Latinoamericana @ *PASCAL*

Entidad principal creada por la Ley de Medios en Argentina para la medición de audiencia (rating) de la televisión. Mi rol al interior de PASCAL era doble: coordinador entre el equipo técnico y el equipo social y administrador de la base de datos y del sistema de encuestas telefónicas automáticas.

2014 Formación para ingreso a doctorado en computación @ *Universidad de Buenos Aires*

2012 – 2013 Unión de Trabajadores de la Economía Popular @ Programa “Argentina Trabaja”

Asistente de quien fue presidente del Instituto Nacional de Tecnología Industrial 2002-2011. Mi tarea estuvo concentrada en las cooperativas de trabajo de “Argentina Trabaja”. Mis tareas estaban dirigidas por Enrique Palmeiro, asistente de Francisco, quien fue designado Papa mientras trabaja con él. Hacíamos una evaluación de las cooperativas para promover su producción.

La sociedad y la cultura vista desde el lente de los sistemas complejos

2011 – 2012 Predicción de alianzas parlamentarias @ *Université de Lausanne*

Curso en el Instituto de Matemática Aplicada de la Universidad de Ciencias Sociales y Políticas de la Universidad de Lausanne sobre complejidad y modelos basados en agentes dictado por Jorge Peña. Tesis de licenciatura donde se implementaba el modelo de Robert Axelrod de predicción de alianzas aplicado a las democracias parlamentarias europeas.

2010 Formación en ciencias exactas y naturales @ *Universidad de Buenos Aires*

2008 – 2009 Antropología, complejidad, simulación y etnografía @ *Grupo Antropocaos*

Asistente de investigación en proyecto UBACyT “Modelos de simulación y antropología, su relación con el trabajo etnográfico”. Docente ayudante en Seminario “Sociedades artificiales y etnografía”.

Educación Universidad de Buenos Aires

2016 – 2022 Doctorado en **Ciencia de la Computación**.

2012 – 2014 Licenciatura en Ciencia de la Computación. (Suspendida por comienzo del doctorado)

2005 – 2009 Licenciatura en **Ciencias Antropológicas**.

Docencia

2024 – actual *Inferencia bayesiana causal* @ *Laboratorios de Métodos Bayesianos - ECyT UNSAM*

(1) Especificación y evaluación de modelos causales alternativos. (2) Inferencia exacta y aproximada (pasaje de mensajes, variacional, MCMC, SBI). (3) Flujo de inferencia, criterio backdoor y Do-Calculus. (4) Algoritmos y frameworks de Causal ML. (5) Algoritmos temporales y modelos temporales generativos. (6) Inferencia contrafactual y efectos causales directos. (7) Toma de decisiones, optimización como inferencia y utilidad ergódica. (8) Intervenciones socioeconómicas e instrumento financiero productivo.

2019 – actual *Mentorias individuales*. Director de tesis de 7 licenciaturas en Computación y Ciencia de Datos

2016 – 2024 Cursos previos en UBA (Computación y Antropología): Inferencia bayesiana, Algoritmos y estructuras de datos (I y II), Introducción a la computación, Programación funcional (`Haskell`), Ciencias sociales computacionales.

Investigación científica

Artículos

○ **Landfried** G., Mocskos E. *TrueSkill Through Time: reliable initial skill estimates and historical comparability in Julia, Python and R*. **Journal of Statistical Software. 2025**. 10.18637/jss.v112.i06

○ **Landfried**, G., Cairo G., Mocskos E. *Network Position and Learning Dynamics: Unveiling the Impact of Social Structure on Skill Acquisition in Online Gaming Platforms*. **Journal of Computational Social Sicence. 2025** 10.1007/s42001-025-00370-2.

○ Longhi, SA; Muñoz-Calderón, A; García-Casares, L; Irazu, L; Rodríguez, MA; **Landfried**, G; Alonso-Padilla, J; Schijman, AG; and Chagas-group *Inter-Laboratory Harmonization Study and Prospective Evaluation of the PURE-T. cruzi-LAMP Assay for Detecting Parasite Presence in Newborn Dried Blood Spots*. **The Journal of Molecular Diagnostics. 2024.** 10.1016/j.jmoldx.2024.08.007

○ Denwood, M; Nielsen, S; Olsen, A; Jones, H; Coffeng, L; **Landfried**, G; Nielsen, M; Levecke, B; Thamsborg, SM; Eusebi, P; Meletis, E; Kostoulas, P; Hartnack, S; Erkosar, B; Toft, N. *All that glitters is not gold: an interpretive framework for diagnostic test evaluation using Ascaris lumbricoides as a conceptual example*. **Plos Neglected Tropical Disease. 2024**.10.1371/journal.pntd.0012481

○ **Landfried**, G; Fernandez Slezak, D; Mocskos, E: *Faithfulness-boost effect: Loyal teammate selection correlates with skill acquisition improvement in online games.* **PLoS one. 2019**. 10.1371/journal.pone.0211014

Habilidades

Si hay equilibrio, menos pesa.

Herramientas

`Delta Lakehouse` (`DataBricks`, `Spark`, `Streaming`, `Pipeline` ...), `Agentic AI` (`Claude Code`, `MCP`, `Hooks`, `Subagents`, `Skills`, ...), `Python` (`Pytorch`, `TensorFlow`, `Keras`, `PyMC`, `Pyro`, `Particles`, `Scipy`, `Sklearn`, `Pandas`, `Numpy`, ...), `R` (`Stan`, `Tidyr`, ...), `Julia` (`Turing`, ...), `C++` (`MPI`), `C#` (`Infer.NET`, ...), `Java`, `Haskell`, `Bash` (`screen`, `ssh`, `vi`, `rsync`, `awk`, ...), `SQL`, `NoSQL`, `Git`, `Docker`, `Latex` (`Tikz`), `Html`, ...

REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Ciencias Exactas y Naturales

El Rector de la Universidad y el Decano de la Facultad

Por cuanto, el Licenciado en Ciencias Antropológicas (orientación Sociocultural) **Gustavo Andres Landfried** natural de la Ciudad de Buenos Aires, ha finalizado el 9 de abril de 2024 los estudios correspondientes al **Doctorado de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Doctor de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Buenos Aires, 20 de noviembre de 2024

Documento Nacional de Identidad 31916400

Título de la tesis: "Análisis bayesiano del aprendizaje en comunidades de video juegos."

Calificación Sobresaliente aprobada el 09/04/2024

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

Documento firmado digitalmente

Firma del interesado

REGISTRADO EN EL LIBRO GENERAL DE GRADOS N° 287 FOLIO 305 CON EL N° 22469

Dirección Nacional de Gestión Universitaria
Departamento de Certificación de Títulos

Landfried, Gustavo Andres
DNI - 31916400
UBA - 294/21

Diploma

Ministerio de Capital Humano
Secretaría de Educación
Subsecretaría de Políticas Universitarias
Argentina

000001693103

REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Filosofía y Letras

El Rector de la Universidad y la Decana de la Facultad

Por cuanto, **Gustavo Andrés Landfried**

natural de la Ciudad de Buenos Aires, ha finalizado el 20 de mayo de 2013 los estudios correspondientes a la carrera de **Ciencias Antropológicas.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Licenciado en Ciencias Antropológicas (orientación Sociocultural).**

Buenos Aires, 12 de agosto de 2014

SOFÍA IRENE THISTED
Secretaria de Asuntos Académicos

GRACIELA ALEJANDRA MORGADE
DECANA

ALBERTO E. BARBIERI
RECTOR

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

Documento Nacional de Identidad 31916400

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MINISTERIO DE EDUCACIÓN

DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

Certifico que las firmas del Rector, Dr. Alberto Edgardo BARBIERI, Secretaria de Asuntos Académicos, Lic. María Catalina NOSIGLIA y el Director General de Títulos y Planes, Sr. Leonardo Federico GOTTIFREDI que anteceden, guardan similitud con la/s que obran en nuestros registros.

Buenos Aires, - 3 NOV 2014

Cont. MARIA I. ILLAPRESCAPE
DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

LANDSRIEDA, GUSTAVO (21)

REPUBLICA ARGENTINA
Ministerio de Educación
Presidencia de la Nación
ME000000620328

Firma del interesado

REGISTRADO EN EL LIBRO GENERAL DE GRADOS Nº 222 FOLIO 184 CON EL Nº 11673

Dirección Nacional de Gestión Universitaria
044081277
Landriaed, Gustavo Andrés
DNI 31916400
UBA - Licenciado en Ciencias Arts

Bayesian Causal Inference

IBC1.2026.1

Professor: Gustavo Landfried[1,2,3]

1. School of Science and Technology, Universidad Nacional de San Martín (UNSAM)
2. Bayesian Methods Laboratory
3. Muttdata

Contact: `glandfried@dc.uba.ar` `glandfried.github.io`

April 13, 2026

All data-driven sciences develop causal arguments to explain and predict the world. The evaluation of causal hypotheses is attracting increasing interest from industry, which needs to measure the real impact of its actions. The advantage of causal models lies in their predictive capacity, which adapts naturally to contextual changes such as human interventions. Moreover, we know that if causal arguments correspond to the underlying causal reality, no artificial intelligence model, however complex, can outperform them.

In this course we will review the foundations for evaluating alternative causal models M given data D, $P(M|D)$, both with and without interventions. We will also review methods for generating predictions of hypotheses H internal to models M, $P(H|D, M)$, both for evaluating out-of-sample causal effects and for predicting the impact of alternative counterfactual actions. Finally, we will address the problem of optimal decision-making as an inference problem in which we verify that none of the alternative counterfactual decisions improve the result on the target variable.

Faced with the dizzying pace of AI tools, this course prioritizes immutable foundations. Despite all advances, since the 18th century no new system for reasoning under uncertainty has been proposed. If machines ever surpass humans, they will have to do science by strictly applying probability rules to evaluate causal theories. To reach (intersubjective) truths under uncertainty, one simply needs to know how to apply and preserve the universal principle of not lying: do not assert more than is known, without hiding what is known. This yields optimal belief distributions given the available information, unimprovable in practical terms.

Objectives

This course focuses on evaluating alternative causal arguments through the (approximation to) strict application of probability rules (Bayesian approach), the system of reasoning under uncertainty. The main objective is to review methods developed in recent decades for:

- **Causal models.** Mathematically specifying causal arguments expressed in natural language through intuitive graphical methods.
- **Inference flow.** Understanding when inference flow opens and closes in causal structures, in order to separate spurious influence from causal influence.
- **Causal effect estimation.** Applying Machine Learning methods to identify the causal effect between pairs of variables from observational data without interventions.
- **Counterfactual predictions.** Specifying joint probability distributions between factual and counterfactual outcomes, and generative reasoning about counterfactual consequences.
- **Decision-making over time.** Selecting optimal decisions in action-perception cycles with a hidden (simulated) nature.

About the instructor

Gustavo Landfried is a professor of Bayesian Causal Inference at the School of Science and Technology of the Universidad Nacional de San Martín (UNSAM), and one of the leading figures in Bayesian causal inference education in Latin America. After obtaining a degree in Social Anthropology, Gustavo earned his PhD in Computer Science from the Universidad de Buenos Aires (UBA) under the supervision of the directors of the Applied Artificial Intelligence Laboratory (Diego Slezak) and the Interdisciplinary High-Performance Computing Laboratory (Esteban Mocskos). He specialized in Bayesian skill analysis for the video game industry, which led him to develop and maintain the state-of-the-art packages in Python, Julia and R (*TrueSkill Through Time*, Journal of Statistical Software, 2025). Gustavo currently works in industry (Muttdata) as a Data Scientist developing Bayesian and causal inference projects for large multinational corporations in Latin America. He is also co-founder and coordinator of the community `bayesplurinacional.org`, organizing annual scientific events across the continent (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). He has supervised seven BSc/MSc theses in Computer Science and Data Science.

Contents

1 Specification and Evaluation of Causal Arguments **3**
1.1 Alternative Causal Arguments and Uncertainty 3
1.2 Surprise: The Problem of Communication with Reality 3

2 Inference Methods and Probabilistic Programming **4**
2.1 Exact Inference and Message Passing 4
2.2 Forgetting: The Problem of Approximate Inference 4

3 Causal Predictions **5**
3.1 Inference Flow and Elimination of Spurious Influence 5
3.2 Estimands and Do-Calculus 5

4 The Algorithm Zoo **5**
4.1 Traditional Methods 5
4.2 Leading Algorithms in Causal Inference Competitions 6

5 Time **6**
5.1 Temporal Algorithms 6
5.2 Generative Temporal Causal Models 6

6 Counterfactuals **7**
6.1 Models with Counterfactuals and Equivalence between Causal Inference Approaches 7
6.2 Using Counterfactuals to Estimate Direct Causal Effects 7

7 Decision-Making **7**
7.1 Decision-Making Over Time: The Epistemic-Evolutionary Cost Function . . . 7
7.2 Temporal Correction of Expected Utility Theory and Optimization as Inference 8

8 Socioeconomic Interventions **8**
8.1 General History, Commons and Successful Institutions 8
8.2 Productive Financial Instrument 8

All sub-units include assessed assignments. Upon completion of each full unit, an evaluation is conducted based on multiple-choice questions with belief distributions.

1 Specification and Evaluation of Causal Arguments

System of reasoning under uncertainty, graphical specification of causal arguments, and evaluation of alternative causal models with and without interventions.

1.1 Alternative Causal Arguments and Uncertainty

Truth and falsehood. Intersubjective agreements: maximum uncertainty given available information. Honest belief distribution across parallel universes of causal arguments. Preservation of prior belief compatible with the data and prediction as a sum of parallel universes compatible with what one wants to predict. System of reasoning under uncertainty: the rules of probability. Graphical methods for mathematically specifying probabilistic causal arguments: Bayesian networks. Basic properties of graphical models: structure, independence and factorization. Measuring uncertainty over hypothesis spaces internal to models, $P(\text{Hypothesis}|\text{Data}, \text{Model})$. Evaluating alternative causal arguments from observational data without interventions, $P(\text{Model}|\text{Data})$. The evidence, $P(\text{Data}|\text{Model})$, and its transformations (geometric mean and logarithm).

Assignment 1.1: Specification and evaluation of Monty Hall and Base models.

1.2 Surprise: The Problem of Communication with Reality

The fundamental elements of the reasoning system under uncertainty (Kolmogorov and Rényi axioms). The invariant structure of empirical data. Its isomorphism with communication systems with reality. Surprise as a source of information. The equivalence between the geometric

mean of the prediction, $P(\text{Data}|\text{Model})$, and cross-entropy. The advantage of causal models that correspond to the underlying causal reality. The mathematical definition of the principle of not lying: maximum entropy given available information. Specification of causal models for contexts with corrupted data. Context-adaptive causal mechanisms: do-operator and gates. Specification of causal arguments through factor graphs. Identification of causal models with and without interventions. Evaluation of alternative measurement procedures.

Assignment 1.2: Specification and evaluation of the Monty Hall model with corrupted data.

2 Inference Methods and Probabilistic Programming

Exact inference methods through message passing between nodes of the causal network, and approximate inference — both analytic and sampling-based — through probabilistic programming languages.

2.1 Exact Inference and Message Passing

Exact inference. The variable elimination algorithm. Decomposition of probability rules as messages exchanged between nodes of causal networks: sum-product algorithm. The extended Monty Hall model case. Conjugate distributions. Beta-Bernoulli model and its applications in medicine and marketing. Univariate Gaussian models and the exact solution to the skill estimation model. Multivariate Gaussian models: posterior and evidence. Bayesian linear regression. Arbitrary hypothesis selection via maximum likelihood. Interpretation of estimated parameters under known causal realities. Evaluation of linear models of increasing complexity. Absence of overfitting through strict application of probability rules. Prediction based on model ensembles, $P(\text{Data})$.

Assignment 2.1: Evaluation of linear models of increasing complexity.

2.2 Forgetting: The Problem of Approximate Inference

Analytic approximation methods. KL divergence and approximate inference via expectation propagation. The case of skill estimation in the video game industry. Reversed KL divergence and approximate inference via variational inference. The Bayesian version of the recommendation system that won the Netflix competition. The probabilistic programming language `Pyro`. Sampling-based approximation methods. Monte Carlo approximate inference: Metropolis-Hastings, Hamiltonian Monte Carlo and Sequential Monte Carlo. Probabilistic programming languages `PyMC`, `Stan`, `Turing`. The case of evaluating alternative measurement instruments and the Bayesian p-value. Generative approximation methods. Approximate Bayesian Computation (ABC): likelihood-free inference via summary statistics and distance metrics. Current methods for estimating the joint distribution via neural networks from simulated data (SBI) and efficient inference.

Assignment 2.2: Evaluation of alternative measurement instruments: Chagas diagnostic test with corrupted data.

3 Causal Predictions

Elimination of spurious influence from predictions and the steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect.

3.1 Inference Flow and Elimination of Spurious Influence

Inference flow in causal networks. Opening and closing of inference flow in basic structures: pipe, fork and collider. Opening and closing of inference flow along paths of any length: d-separation. Elimination of spurious influence from predictions: backdoor controls. Good and bad controls. Neutral controls and heterogeneous causal effects. Use of linear regression for estimating non-linear causal effects. Interpretation of estimated coefficients. The Table 2 Fallacy. The steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect. Discovery of causal structures compatible with estimated joint probability distributions from data.

Assignment 3.1: Causal predictions under diverse underlying causal realities.

3.2 Estimands and Do-Calculus

The estimand for the general causal effect: adjustment formula and propensity score. The fundamental characteristic of randomized experiments: independence between treatment and its potential counterfactual outcomes. The "conditional ignorability" analogy as a criterion for selecting control variables in the potential outcomes approach to causal inference. Estimand of causal effects under conditional ignorability and deterministic consistency. The 3 rules of Do-Calculus: Rule 1, Ignoring an observation; Rule 2, Intervention as observation; Rule 3, Ignoring an intervention. Alternative estimand when hidden controls exist: frontdoor formula. Alternative estimand when randomized experiments fail: instrumental variables.

Assignment 3.2: Estimation of causal effects: general, frontdoor and instrumental.

4 The Algorithm Zoo

From early strategies to state-of-the-art supervised learning models with proven capacity for estimating heterogeneous causal effects in high-dimensional settings.

4.1 Traditional Methods

Traditional effect estimation methods. Comparison of treated and control units with similar characteristics: Matching. Use of the Propensity Score as a similarity measure. Reweighting the sample through the Propensity Score: Inverse Probability Weighting (IPW). Limitations of traditional methods in high-dimensional settings. The Causal Machine Learning (CausalML) solution: high-capacity predictive algorithms for estimating heterogeneous causal effects (CATE). The Python causal inference ecosystem: integration of `DoWhy`, `EconML` and `CausalML`. The concept of Meta-Learners: strategies for decomposing estimation of the conditional causal effect (CATE) into standard supervised prediction problems. S-Learner and T-Learner: advantages and disadvantages. The impact of lack of "positivity" on causal predictions: the personal lending industry case.

Assignment 4.1: Causal predictions under different degrees of positivity.

4.2 Leading Algorithms in Causal Inference Competitions

Fundamentals and application of the Doubly Robust Learner. Causal inference competitions at the Atlantic Causal Inference Conference. The regularization problem in causal effect estimation. Estimation strategy under partially linear models: orthogonalization. Fundamentals and application of Doubly Machine Learning: regularization in nuisance models and isolation of the causal effect. Non-linear algorithms based on tree ensembles. From Random Forests to Causal Forests. From Bayesian Additive Random Trees to Bayesian Causal Forests. The problem of estimated confidence intervals in data competitions and possible solutions. Hybrid algorithms.

Assignment 4.2: Causal inference competition ACIC 2022.

5 Time

Specialized algorithms for estimating causal effects in time series. Generative temporal causal models: specification, inference and evaluation.

5.1 Temporal Algorithms

The advantage of longitudinal data: time-invariant fixed effects. Classical Difference-in-Differences (DiD). The problem of DiD with heterogeneous effects. DiD with multiple periods and staggered adoption. New robust estimators: Callaway-Sant'Anna, Sun-Abraham, Doubly Robust DiD. The problem of absence of comparable control units. Construction of synthetic control groups through weighted combination of untreated units. Combining synthetic weights with DiD structure allowing staggered timing and heterogeneous trends: Synthetic Difference-in-Differences. Generalization of synthetic controls for multiple simultaneously treated units via missing data imputation: Matrix Completion.

Assignment 5.1: Estimation of temporal causal effects comparing different methods.

5.2 Generative Temporal Causal Models

Specification of temporal causal models and their unrolling. Skill estimation in the video game and sports industries. Inference flow in complete-history generative models: forward and backward propagation. Undefined messages and convergence via Loopy Belief Propagation. The forward-backward algorithm (smoothing): skill estimation over time. Temporal predictions and leave-one-out predictions: costs. Skill estimation in the history of tennis. Interpretation of odds offered by bookmakers. Simplified betting game: the honest coin. Properties of betting games. The advantage of individual diversification (negative entropy).

Assignment 5.2: Specification of a generative temporal causal betting model. Simulation and impact analysis of cooperation and individual saving.

6 Counterfactuals

Specification of models with counterfactuals, equivalence between causal inference approaches, and decomposition of total effects into natural direct and indirect effects via the mediation formula.

6.1 Models with Counterfactuals and Equivalence between Causal Inference Approaches

The levels of causal reasoning. Specification of causal models with counterfactuals: Twin Networks. Prediction of counterfactuals using factual information: Monty Hall. Revisiting the equivalence of the two causal inference approaches: Potential Outcomes and Structural Causal Models. Compliance analysis of ignorability via Twin Networks: examples. Specific constraints of Structural Causal Models: one exogenous variable per endogenous variable and determinism of endogenous variables. Breaking of Ignorability-Backdoor equivalence in general causal generative models: the paradigmatic counterexample. Generalization of the Backdoor causal inference approach to general causal generative models.

Assignment 6.1: Predicting counterfactual values using factual information.

6.2 Using Counterfactuals to Estimate Direct Causal Effects

Total vs. Direct and Indirect causal effects. Conditional Direct Effect: a contraindicated intervention. Decomposition of the total effect: Natural Direct Effect, the impact produced by treatment under an untreated mediator; Natural Indirect Effect, the impact produced by the change in the mediator under treatment. The Burks-Hammel debate: measuring gender discrimination in university admissions. Identifiability of natural effects: elimination of spurious influence between the mediator and the outcome. Mediation formula: NDE and NIE estimands from observational data and structural models. Estimation of NDE and NIE from observational data without interventions.

Assignment 6.2: Evaluation of discrimination in simulated data.

7 Decision-Making

The probability-evolution isomorphism, consequences of the multiplicative nature of variant selection, and the temporal correction of expected utility theory.

7.1 Decision-Making Over Time: The Epistemic-Evolutionary Cost Function

Von Neumann's expected utility theory. John L. Kelly's critique of Von Neumann. Betting games. The structural equivalence between betting games, hypothesis evaluation in probability, and natural selection in evolution. The advantage of individual diversification (negative entropy). The emergent properties of material forms and the advantage of variance reduction through cooperation. The fallacy of the prisoner's dilemma and the tragedy of the commons. The advantage of cooperative specialization. The Kelly criterion: cooperating with oneself. Fractional Kelly.

Assignment 7.1: Skill estimation and tennis betting.

7.2 Temporal Correction of Expected Utility Theory and Optimization as Inference

Systems of interaction with reality: action-perception. Active learning. Prediction by maximum uncertainty and action selection by minimum uncertainty. The ergodic (temporal) correction of expected utility theory. Optimal factual decision as absence of superior counterfactual alternatives. Optimization as inference. Counterfactual-based influence diagrams: the betting game and solution to specialization (individual and 2-person pool). Deterministic policies and random drip experiments. The personal lending industry case: deterministic credit allocation. Strategies for conducting micro-randomized experiments, minimizing data production and collection: the value of information.

Assignment 7.2: Micro-experiments and policy evaluation in the personal lending industry.

8 Socioeconomic Interventions

History of the colonial-modern turn, governance of the commons, analysis of current cases, and design of productive financial instruments at all levels (savings, land use planning, basic services, housing access, cycle reproduction).

8.1 General History, Commons and Successful Institutions

Limits of estimating causal effects between pairs of variables. The case of gender violence and random allocation of housing subsidies. The development of culture and agriculture. The 6 innovation centers of humanity. The colonial-modern turn: feudal pandemic, American silver, and Chinese technology. The era of genocides and the loss of cultural diversity. The deepening of the current ecological crisis and the inability of modern metropolitan science to resolve it. Characteristics of successful institutional systems in the management of the commons. Egypt: land distribution, agricultural productivity, population density and demographic dynamics. Argentina: land distribution, agricultural productivity, population density and demographic dynamics.

Assignment 8.1: Report: analysis of Argentina and Egypt data; estimates, conclusions and action proposals.

8.2 Productive Financial Instrument

Financial-productive systems in present-day Argentina: interest rate, risk and liquidity. Source of real value: water, land and life. Productive savings financial instrument — sowing pool — based on intensive agriculture and rurban development. The three actors of the system: savers, residents and administration. Savings based on pools for purchasing rural land (blocks of more than 100 ha) and interest rates aligned with rural rental values. General land planning: roads, housing, market-garden agriculture and common spaces. Specific planning of sectors and minimum size of rural areas (plots of 2000 m^2) and dwellings (250 m^2). Development of basic services: water, sewage, foundations and common area. Requirements for residents to acquire ownership: total cost and timelines for debt cancellation. The

complete cycle for process reproduction. Estimated capabilities and speed of potential growth.

Assignment 8.2: Writing the technical report and executive summary of the productive financial instrument.

Core Bibliography

- Koller, D. & Friedman, N. (2009). *Probabilistic Graphical Models.* MIT Press.
- MacKay, D. (2003). *Information Theory, Inference and Learning Algorithms.* Cambridge University Press.
- Jaynes, E.T. (2003). *Probability Theory: The Logic of Science.* Cambridge University Press.
- Pearl, J. (2009). *Causality: Models, Reasoning and Inference.* Cambridge University Press.
- Pearl, J. & Mackenzie, D. (2018). *The Book of Why.* Basic Books.
- Peters, J., Janzing, D. & Schölkopf, B. (2017). *Elements of Causal Inference.* MIT Press.
- Bishop, C. (2006). *Pattern Recognition and Machine Learning.* Springer.
- McElreath, R. (2020). *Statistical Rethinking.* CRC Press.
- Molak, A. (2023). *Causal Inference and Discovery in Python.* Packt.

Probabilistic Programming Languages

- `Pyro` (Uber AI): https://pyro.ai
- `PyMC`: https://www.pymc.io
- `Stan`: https://mc-stan.org
- `Turing.jl` (Julia): https://turinglang.org